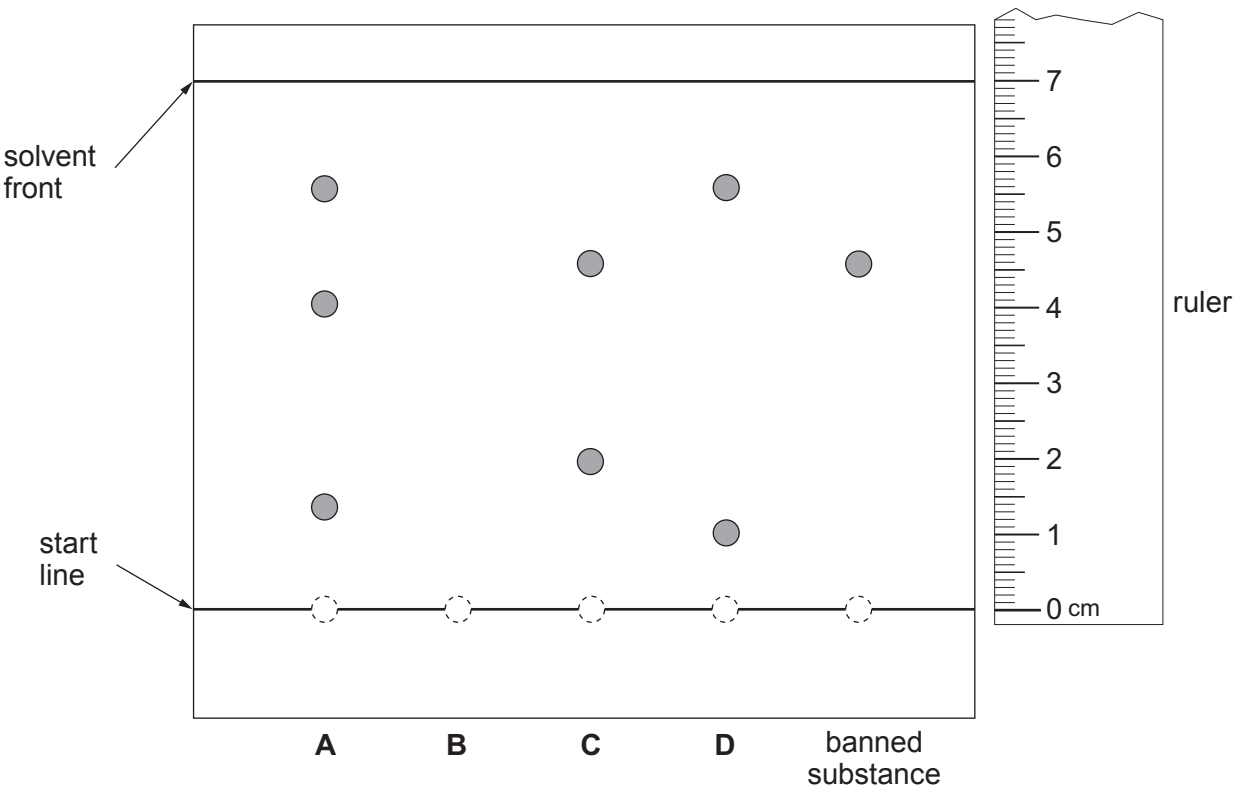


Examiner
only

2. A food company was accused of using a banned substance in its sweets. Scientists tested four dyes, **A-D**, to find out if this was true or not. The results are shown below.



(a) Name the method used. [1]

.....

(b) Dye **B** was found to have an R_f value of 0.428.

(i) Use the following equation to calculate the distance moved by dye **B**. [2]

distance moved by dye **B** = $R_f \times$ distance moved by solvent front

Distance moved = cm

(ii) **Complete the diagram** above to show the position of dye **B**. [1]



Examiner
only

(c) Identify which dye, **A-D**, contains the banned substance. Give a reason for your answer. [2]

Dye

Reason

.....

6

3410U101
05

